

JACOB YOON

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EXPERIENCE

Epic Systems

Software Developer Intern

Verona, WI

May 2025 – Aug 2025

- Designed an **LLM**-powered conversational provider search assistant for public hospital websites
- Engineered a cloud-based, multi-agent workflow using internal **.NET** frameworks and **Azure** infrastructure to support real-time, high-volume queries
- Developed simulation and evaluation frameworks to assess assistant behavior across **100+** conversation scenarios
- Implemented interactive **React** components for displaying rich provider results within the chat interface
- Created **LLM** evaluation pipeline to benchmark the impact of organization-specific knowledge base **RAG** on assistant performance over **1,500** simulated patient queries

Dassault Systèmes

Software Engineer Intern

Rochester, NY

Jun 2024 – Dec 2024

- Built full-stack automation platform integrating Jira with internal issue tracking systems using **Angular** and **Node.js**, streamlining cross-functional workflows and saving **15+ hours** of weekly manual work
- Delivered critical feature enhancements and bug fixes for 3DEXPERIENCE platform using **.NET**, **Angular**, and **PostgreSQL**, supporting **500+** enterprise users in manufacturing workflows

Dassault Systèmes

Software Engineer Intern

Ithaca, NY

May 2023 – Dec 2023

- Overhauled legacy search architecture using **Angular** and **.NET**, achieving **90%+** performance improvement
- Optimized database query performance through **.NET** endpoint refactoring, reducing response times by **60%**
- Developed internal tools with **Angular**, **Express**, and **Docker** to automate data generation for app environments

EDUCATION

Rochester Institute of Technology

Bachelor of Science and Master of Science in Computer Science

Rochester, NY

Aug 2021 – May 2026

- GPA: **4.0/4.0** (MS), **3.89/4.0** (BS) | **Dean's List** (All Semesters)
- Relevant Coursework: Information Retrieval, Computer Vision, Machine Learning, AI, Computational Geometry, Graph Databases, Big Data, Parallel and Distributed Systems, Algorithms

PUBLICATIONS

Yoon, J. (2026). *MathAMR+: A Unified Graph Neural Network Framework for Multimodal Mathematical Information Retrieval* [Master's thesis]. Rochester Institute of Technology. <https://www.proquest.com/docview/3336734205> [Published].

Yoon, J., & Zanibbi, R. (2026). Searching for Solutions: A Survey of Mathematical Reasoning with Large Language Models. *ACM Computing Surveys*. [Submitted].

PROJECTS

Audio Spheres

audio-spheres.web.app | github.com/jyoon926/audio-spheres

Nov 2024 - Dec 2024

- Built a music discovery platform using **React** and **Spotify Web API**, visualizing user music taste through an interactive, tree-based interface with pan and zoom navigation
- Implemented **Firebase** cloud persistence and **CI/CD** pipeline for automatic deployment

SKILLS

Languages: Python, TypeScript/JavaScript, C#, Java, C, C++

Frameworks/Libraries: React, Angular, .NET, Next.js, Node.js, Express, Spring, Tailwind CSS

AI/ML: PyTorch, PyTerrier, Hugging Face, NumPy, Pandas

Databases/Tools: Supabase, Firebase, MySQL, PostgreSQL, MariaDB, AWS, MongoDB, Docker, Jenkins